AutoDoomscroller: An On-Device Agent That Learns Which Short Videos You Would Skip

Project Proposal

Juyong Park
Business Administration
Seoul National University
Republic of Korea
tony3ynot@snu.ac.kr

ABSTRACT

Short-video feeds are ranked to keep people watching, yet in six days of our own logs 38% of the non-ad reels were swiped away within two seconds. We propose AutoDoomscroller, an Android agent that learns from one person's scrolling which Instagram Reels and YouTube Shorts they would skip, then skips them, lets them scroll and like with gestures, and shows them what they watch. Everything runs on the phone. A prototype already records both apps, and a first model reaches a skip AUC of 0.63.

1 MOTIVATION

Scrolling a short-video feed means making the same small decision hundreds of times a day: watch this, or swipe it away. In six days our prototype logged 1,833 reels. Of the non-ad reels, 38% were left within two seconds, half of those within about one second (Fig. 1). The feed itself is chosen by the platform's recommender, which is built to maximise watch time [1, 11], and users of such feeds report a weakened sense of agency [6].

We ask whether the phone can learn those quick judgements from the user's own behaviour and act on them: skip what the user would skip anyway, let them move through the feed without touching it, and show them where their time goes. Unlike the platform's recommender, such an agent works for the user and keeps their data on the device. Building it touches every part of a mobile ML system: sensing, on-device learning, model compression and visualization.

2 LITERATURE REVIEW

Platforms predict engagement from the logs of millions of users: YouTube ranks videos by expected watch time [1], Kuaishou corrects watch-time prediction for video length [11], and dwell time is a well-established signal of preference [10]. EdgeRec even moves part of the ranking onto the phone so that it reacts to the user's latest actions [2], but it still serves the platform's goals. We instead predict one person's skips on their own phone, from a few hundred examples and without access to the platform.

Digital self-control tools mostly block or limit use [7], and Lukoff et al. found that autoplay and recommendations undermine users' sense of agency [6]. Our agent does not block anything; it acts on the user's own past choices and makes them visible.

For hands-free control, EyePhone showed early on that a phone can be operated with eye blinks [8], and MediaPipe now runs face and hand landmark models in real time on phones [5]. For the models we use MobileNetV3 on frames [3] and multilingual E5 on text [9], and compress them with integer quantization [4] and pruning.

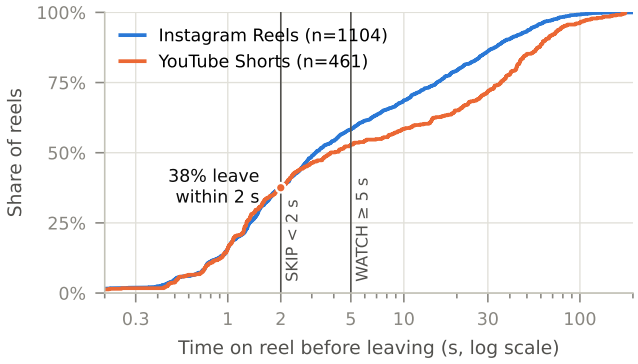


Figure 1: Time spent on each non-ad reel before swiping away (six days, one user). In both apps 38% of reels are left within 2 s.

3 PROPOSED DESIGN

Sense (Fig. 2). An accessibility service reads the screen of Instagram Reels and YouTube Shorts a few times per second to know which reel is showing, how long it stayed, and whether the user liked, saved or commented on it or watched it to the end. Once a reel has settled after the swipe, the service takes one screenshot cropped to the reel. In hands-free mode, the front camera is used for face and hand tracking.

Label. A reel counts as SKIP if the user left within 2 s, and as WATCH if they interacted with it, watched it to the end, or stayed at least 5 s. Everything in between is left out. Likes alone would not work as a target: there were only 6 in 1,833 reels.

Learn. The model may only use what is known when a reel appears: the first frame, the text on it (read with on-device OCR such as ML Kit), the caption, and the creator's history. For each of these we look up the most similar reels the user has already seen and use the share of skips among them as a score; a logistic regression combines the scores into $P(\text{skip})$. Because the scores come from a growing memory, the model keeps improving without retraining the encoders.

Act. Auto mode swipes when $P(\text{skip})$ is high enough to be right 90% of the time; if the user goes back to an auto-skipped reel, that counts as a correction. Hands-free mode maps a double wink to

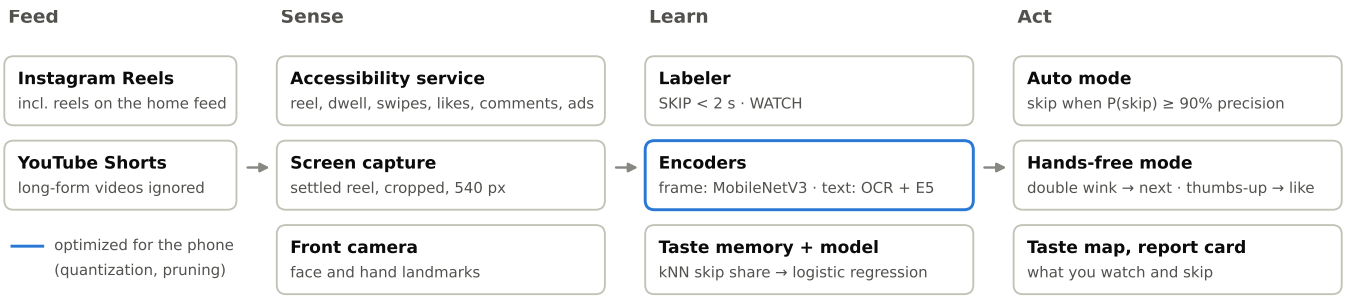


Figure 2: System overview. Everything runs on the phone.

Table 1: Skip AUC on 933 labeled reels, each day predicted from the days before it (mean, 95% CI).

Inputs	Skip AUC
Creator history	0.63 (0.58–0.68)
First frame (MobileNetV3)	0.56 (0.52–0.61)
Text on the frame (OCR)	0.58 (0.53–0.62)
Creator + frame + caption	0.62 (0.57–0.67)
+ session context	0.56 (0.52–0.61)

"next" and a thumbs-up to "like". A taste map and a per-session report card show the user's habits.

4 PROTOTYPE IMPLEMENTATION

The Android app (Kotlin, tested on a Galaxy S24) already tracks both apps, detects and skips ads, and exports its logs for offline analysis in Python, where we compute embeddings, run OCR and evaluate models on days they were not trained on.

Preliminary results. Table 1 shows the first results. The creator's history alone reaches an AUC of 0.63; adding the frame and caption does not help yet. The most promising content signal is the text on the frame: on reels where a sharper 540 px frame is available, its AUC rises from 0.46 to 0.63, so the app now captures that frame first. Information about the session, such as time of day and recent scrolling pace, makes things worse, because the user's mood changes from one session to the next.

5 POTENTIAL CHALLENGES

- *Weak signal.* First impressions are noisy and depend on mood. Sharper on-screen text, per-session thresholds and online updates should help.
- *Speed and battery.* Half of the quick skips happen within about a second, so capturing, reading and scoring a reel must take a few hundred milliseconds, and the screenshots and camera must not drain the battery.
- *Measuring an agent that acts.* Auto-skipped reels lose their label, so a random 10% of predicted skips will be let through to measure precision.
- *Fragility.* The app depends on the apps' screen layouts (partly on Korean UI text), and the data comes from one user.
- *Privacy.* Screenshots and camera frames never leave the phone.

6 EXPECTED OUTCOMES AND PLAN

(1) An Android app with Learning, Auto and hands-free modes, with all inference on the phone. (2) A skip model with an AUC of at least 0.70 on unseen days and at least 90% precision in Auto mode. (3) A compression study of the encoders (FP32, FP16, INT8 quantization, pruning) reporting size, on-device latency and accuracy before and after. (4) Battery use per mode, a taste map and a report card.

Plan. Nov. 1–7: proposal; collect data with the new capture. Nov. 8–14: hands-free mode. Nov. 15–21: better model; run it on the phone. Nov. 22–28: compression study; Auto mode. Nov. 29–Dec. 5: measurements, visualization, final report.

REFERENCES

[1] Paul Covington, Jay Adams, and Emre Sargin. 2016. Deep Neural Networks for YouTube Recommendations. In *Proceedings of the 10th ACM Conference on Recommender Systems (RecSys '16)*. 191–198. https://doi.org/10.1145/2959100.2959190

[2] Yu Gong, Ziwen Jiang, et al. 2020. EdgeRec: Recommender System on Edge in Mobile Taobao. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM '20)*. 2477–2484. https://doi.org/10.1145/3340531.3412700

[3] Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, Quoc V. Le, and Hartwig Adam. 2019. Searching for MobileNetV3. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 1314–1324.

[4] Benoit Jacob, Skirmantas Kligys, Bo Chen, Menglong Zhu, Matthew Tang, Andrew Howard, Hartwig Adam, and Dmitry Kalenichenko. 2018. Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2704–2713.

[5] Camillo Lugaresi et al. 2019. MediaPipe: A Framework for Building Perception Pipelines. arXiv:1906.08172

[6] Kai Lukoff et al. 2021. How the Design of YouTube Influences User Sense of Agency. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. https://doi.org/10.1145/3411764.3445467

[7] Ulrik Lyngs, Kai Lukoff, Petr Slovák, Reuben Binns, Adam Slack, Michael Inzlicht, Max Van Kleek, and Nigel Shadbolt. 2019. Self-Control in Cyberspace: Applying Dual Systems Theory to a Review of Digital Self-Control Tools. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. 131:1–131:18. https://doi.org/10.1145/3290605.3300361

[8] Emiliano Miluzzo, Tianyu Wang, and Andrew T. Campbell. 2010. EyePhone: Activating Mobile Phones with Your Eyes. In *Proceedings of the Second ACM SIGCOMM Workshop on Networking, Systems, and Applications on Mobile Handhelds (MobiHeld '10)*.

[9] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual E5 Text Embeddings: A Technical Report. arXiv:2402.05672

[10] Xing Yi, Liangjie Hong, Erheng Zhong, Nathan Nan Liu, and Suju Rajan. 2014. Beyond Clicks: Dwell Time for Personalization. In *Proceedings of the 8th ACM Conference on Recommender Systems (RecSys '14)*.

[11] Ruohan Zhan, Changhua Pei, et al. 2022. Deconfounding Duration Bias in Watch-time Prediction for Video Recommendation. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22)*. arXiv:2206.06003